

ZAINAB IRFAN

AI/ML Engineer | Python Developer | FastAPI, RAG & LLM Systems Specialist

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PROFESSIONAL SUMMARY

Software Engineering undergraduate and AI/ML engineer specializing in Retrieval-Augmented Generation (RAG), LLM application development, and full-stack Python systems. Built and deployed multiple production-style AI products spanning conversational agents, document intelligence, clinical data analytics, and workflow automation, using FastAPI, Next.js, LangChain, and vector databases such as Qdrant. Holds 7 industry certifications from AWS, IBM, and DeepLearning.AI.

TECHNICAL SKILLS

Languages	Python, JavaScript, HTML, CSS
Frameworks	FastAPI, Streamlit, Next.js, React
AI / LLM Engineering	RAG Pipelines, LangChain, Prompt Engineering, LLM Evaluation & Benchmarking, Conversational AI Agents, Streaming Chat Interfaces,n8n
Machine Learning	Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation, Bias Detection, Exploratory Data Analysis (EDA)
Data & Vector Stores	Qdrant, Supabase, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
APIs & Cloud	REST API Design & Integration, AWS (AI/ML), Render Deployment, Infrastructure Automation
Languages Spoken	English (Proficient), Urdu (Native)

PROJECTS

Prism AI — RAG-Based Document Intelligence Platform2026

Stack: Next.js, FastAPI, LangChain, Qdrant, Supabase, OpenAI/Claude APIs · Deployed on Render

- Built a full-stack Retrieval-Augmented Generation platform that ingests multi-format documents (PDF, DOCX, PPT, CSV) and returns structured, context-aware answers.
- Designed the retrieval pipeline with LangChain and Qdrant for semantic search, using Supabase for persistent storage of documents and chat history.
- Integrated LLM APIs (OpenAI and Claude) with streamed chat responses for a real-time conversational Q&A experience.
- Built an analytics dashboard visualizing usage insights — queries per day, question volume, and engagement trends — with interactive charts.
- Deployed a production build with a Next.js frontend and a FastAPI backend hosted on Render.

AI Customer Support Agent for Shopify Stores2025–2026

Stack: Python, LLM APIs, Shopify integration

- Developed an AI-powered customer support agent that integrates with a Shopify storefront to handle customer queries automatically.
- Used LLM APIs to generate context-aware responses, reducing manual reply workload for store operations.

Automated Email Reply System2025

Stack: Python, LLM APIs, Email Automation,n8n

- Built an automation system that reads incoming emails, generates context-appropriate draft replies using an LLM, and sends responses automatically.

Comprehensive Clinical Data Analysis & Visualization2025

Stack: Python, Pandas, Matplotlib/Seaborn, Scikit-learn

- Performed end-to-end exploratory data analysis (EDA) on raw clinical datasets, including cleaning, transformation, and feature engineering.
- Built visualizations of clinical insights such as patient distribution by disease and other key trends across the dataset.

- Applied machine learning techniques to generate predictive insights from patient records.

AI PDF Question-Answering Platform2025

Stack: Python, FastAPI, LLM APIs

- Built a tool that allows users to upload a PDF and ask natural-language questions, returning direct answers sourced from the document.

LLM Platform Evaluation & BenchmarkingIn Progress, 2026

Stack: Python, LLM APIs, Open source models, Hugging face models, model integration in system

- Currently developing a benchmarking framework to evaluate and compare the performance of different LLM platforms across defined tasks and metrics.

Machine Learning Models Development2024–2025

- Trained 5+ classification and regression models with a full preprocessing pipeline: encoding, normalization, scaling, and train/test splitting.
- Evaluated models using accuracy, F1-score, precision/recall, and confusion matrices; selected optimal algorithms per dataset characteristics.

RESTful API Development & ML Integration2024

- Engineered FastAPI-based REST APIs (GET/POST/PUT/DELETE) with request validation, integrating trained ML models as live prediction endpoints.

CERTIFICATIONS

- AI Fundamentals & Cloud — AWS
- AI Workflow: Feature Engineering & Bias Detection — IBM
- Building Systems with the ChatGPT API — DeepLearning.AI
- Machine Learning for All — University of London
- Machine Learning — Simplilearn
- App Deployment & Security — Packt
- Learn to Code with AI — Scrimba

EDUCATION

Bachelor of Science — Software Engineering

Pakistan · In Progress

Relevant coursework: Machine Learning, Data Structures & Algorithms, Software Engineering, Web Development